

Agent-based fire evacuation model using social learning theory and intelligent optimization algorithms

Peng Lu^{a,b,c,d}, Yufei Li^{b,c,*}

^a Department of Sociology, Central South University, Changsha 410083, China

^b School of Automation, Central South University, Changsha 410083, China

^c PKU-Wuhan Institute for Artificial Intelligence, Wuhan 430072, China

^d School of Economics and Management, Shaanxi University of Science and Technology, Xi'an, 710016, China

ARTICLE INFO

Keywords:

Fire evacuation
Agent based model
Social knowledge
Random forest
Decision optimization

ABSTRACT

Fire incidents often lead to a series of social problems. Therefore, it is particularly important to optimize evacuation strategies and promote relevant social safety knowledge. Based on this, the study proposes a fire evacuation model that integrates the Fire Dynamics Simulator (FDS) with Agent-Based Modeling (ABM) to simulate a bar fire scenario. In this model, the concept of social learning is introduced, and multiple factors such as evacuation time, trampling risk, and pedestrian health are considered as risk evaluation indicators. Machine learning combined with intelligent optimization methods is applied to optimize evacuation strategies. **First**, we validate the effectiveness of the model by comparing the averaged simulation results with real-world data. The results demonstrate that the simulation outcomes of our model exhibit good accuracy and robustness. **Secondly**, we analyze the importance of the second-floor safety exit. When the second-floor safety exit remains unobstructed, evacuation efficiency and casualty risk can be significantly improved. **Then**, we examine the role of social knowledge. When people are aware of the fire risk and choose to evacuate immediately, casualties can be significantly reduced. **Finally**, we study the effectiveness of phased evacuation in enhancing crowd safety. By employing a method that combines Random Forest and the Particle Swarm Optimization-Genetic Algorithm (PSO-GA), phased evacuation strategies are optimized, resulting in definitive strategies to reduce evacuation risks. This finding further expands social knowledge, indicating that when the proportion of staggered evacuation is appropriate, evacuation risks can be significantly reduced. Our research contributes to the development of social safety knowledge and provides methodological references for formulating evacuation strategies in different settings.

1. Introduction

Fire incidents not only cause significant loss of life and property but also trigger a series of social problems, exacerbating societal crises [1–3]. Therefore, strengthening fire prevention and response measures, along with raising public safety awareness, is crucial for maintaining social stability and promoting economic development [4]. Given the profound societal impacts of fire incidents, as discussed above, it is crucial to develop robust fire risk assessment methods to better understand and mitigate these risks [5–6]. Hu et al. (2022) conducted an in-depth analysis of fire risk assessment in China's informal settlements (IS), which is essential for effective IS management and ensuring environmental safety and public health. Their study categorized various IS

types in China and evaluated the fire risks of 26 buildings within these settlements using risk indices and Bayesian networks to provide semi-quantitative and scenario-based fire risk insights [7]. Landucci et al. (2015) proposed an innovative method to assess the effectiveness of safety barriers, focusing on their role and feasibility in mitigating fire-induced domino effects [8]. These research efforts enhance our understanding and awareness of fire-related risks.

An essential aspect of fire risk assessment is evaluating the safety of human life during a fire. Therefore, studying crowd behavior and the interaction between people and their environment during fire evacuations is critical importance. Current fire evacuation research primarily encompasses five areas: real-life evacuation experiments [9–12], analysis of past fire incidents and improvement of evacuation strategies

* Corresponding author.

E-mail address: liyufei@whai.pku.edu.cn (Y. Li).

<https://doi.org/10.1016/j.ress.2025.111000>

Received 31 July 2024; Received in revised form 31 January 2025; Accepted 3 March 2025

Available online 4 March 2025

0951-8320/© 2025 Elsevier Ltd. All rights reserved, including those for text and data mining, AI training, and similar technologies.

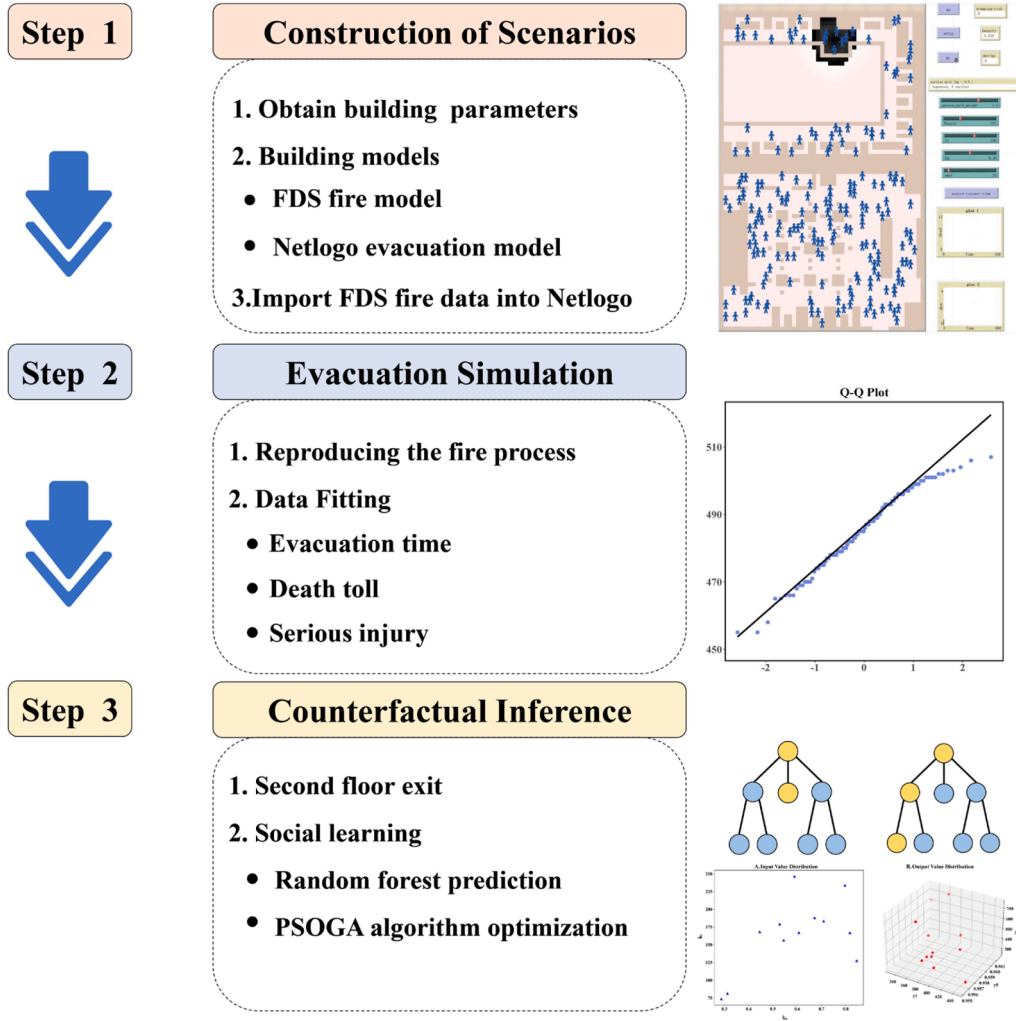


Fig. 1. Research Step Framework. First, extract building parameters to establish a fire evacuation model for crowds, then validate the model's effectiveness through numerical simulation, and finally conduct counterfactual deduction to optimize evacuation strategies.

[13–16], application of advanced intelligent technologies [17–19], evacuation path optimization [20–24], and crowd movement modeling and simulation [25–28]. Although there are numerous methods and substantial research on fire evacuation, modeling and simulation technologies possess distinct advantages in fully replicating the conditions and dynamics of actual fire incidents [29,30]. Evacuation models primarily include social force models, cellular automaton (CA) models, and agent-based models (ABM), which can accurately describe individual behaviors [31]. However, the Social Force Model and Cellular Automata have limitations in simulating complex individual behaviors and social interactions. The former struggles with handling intricate decision-making and heterogeneous behaviors, while the latter, due to its simple rules and strong homogeneity, has difficulty capturing long-range interactions and dynamic adaptability [32,33].

In contrast, Agent-Based Modeling (ABM) offers distinct advantages in studying fire evacuation, making it a powerful tool for analyzing and predicting individual and interactive behaviors in emergency situations [34]. **(a) Detailed simulation of individual behaviors:** ABM allows for the simulation of each evacuee's (agent's) behavior, considering their unique attributes such as age, physical condition, and psychological state [35]. **(b) Capturing complex interactions:** Group dynamics during fire evacuations are complex, involving competition, cooperation, and group influence. ABM can simulate these intricate interactions, such as following behavior, crowding, and queuing phenomena during evacuation, to better understand the impact of different behavior

patterns on evacuation efficiency [36]. **(c) Flexibility and scalability:** ABM models can flexibly add or modify various environmental variables and agent behavior rules to accommodate different research needs and scenario settings. This scalability allows researchers to incorporate more real-world conditions into the model, such as changing environmental factors and different emergency response strategies [37]. Meanwhile, previous fire evacuation simulation studies have seldom employed numerical fitting with actual fire incidents to validate the reliability of their models or to replicate certain collective behaviors observed in real fire situations. Moreover, in the field of intelligent optimization algorithms, prior research has primarily focused on optimizing evacuation routes [20–24]. There has been limited comprehensive assessment of fire risk, and few studies have utilized intelligent optimization algorithms to enhance evacuation strategies aimed at reducing overall fire risk. Additionally, fire evacuation has rarely been studied from the perspectives of social knowledge and social learning.

Based on this, the study proposes a fire evacuation model that integrates Fire Dynamics Simulator (FDS) and ABM to replicate a bar fire scenario in Taiwan [38,39]. In this model, the concept of social learning is introduced, with multiple factors such as evacuation time, trampling risk, and pedestrian health considered as risk evaluation indicators. Machine learning, combined with intelligent optimization methods is applied to optimize evacuation strategies. First, we combine optimal solutions with counterfactuals to form social knowledge [40]. Then, we study the impact of the proportion of people with delayed evacuation

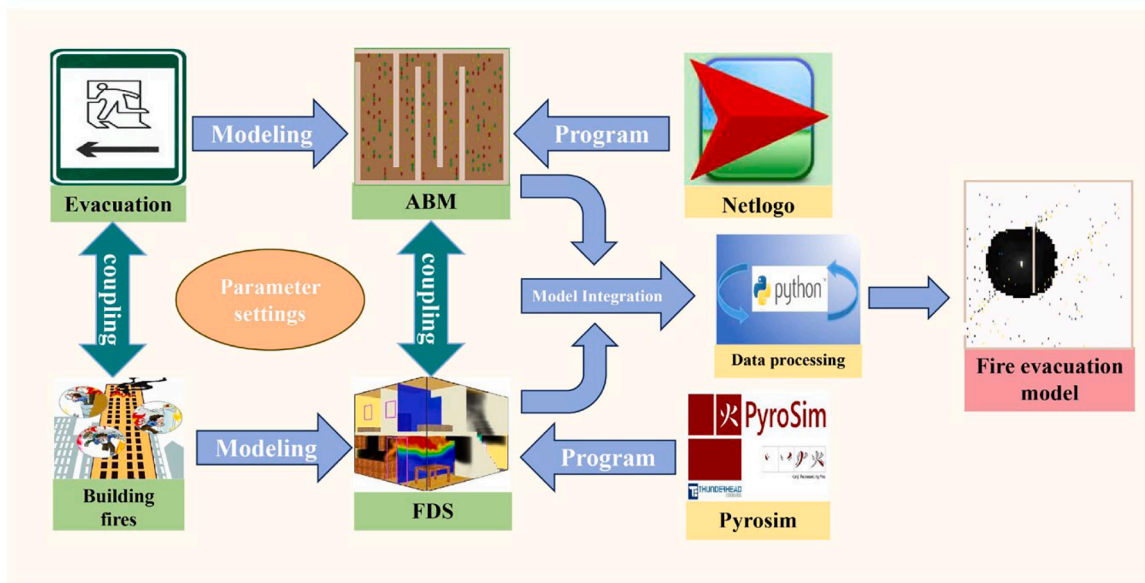


Fig. 2. Schematic diagram of our coupled model. This dynamic coupling allows for the integration of disaster data with crowd movement behaviors at different time steps.

and the delay time on fire risk. By using a method that combines Random Forest with PSO-GA intelligent optimization, we select the optimal evacuation strategy to further reduce fire risk [41]. The primary aim of this study is to derive social safety knowledge through counterfactual simulations based on actual fire incidents. Additionally, the study seeks to identify optimal evacuation strategies using intelligent optimization algorithms. This method can be applied to model fire evacuation scenarios in various public spaces. By identifying optimal evacuation strategies under varying crowd densities, effective emergency plans can be developed to further enhance public safety. The specific technical process is shown in Fig. 1.

The paper is organized as follows. In Section 2, we outline the fundamental principles of the methods used in the model. In Section 3, we describe the relevant information of the fire scenario and conduct modeling simulations to replicate the fire. Additionally, we fit the model to casualty data to validate its effectiveness. In Section 4, we perform counterfactual simulations to derive social knowledge related to public safety and apply intelligent optimization algorithms to optimize evacuation strategies, identifying the optimal evacuation plan.

2. Methods and materials

2.1. Real target case

At approximately 1:15 AM on March 6, 2011, a fire erupted at the Ala nightclub in Taichung City. The nightclub was housed in a two-story building situated in a narrow alley within a residential area (https://news.sohu.com/20110307/n279694812_2.shtml). The fire was ignited by fireworks attached to LED sticks, which were being waved by dancers employed by the nightclub. These fireworks accidentally set fire to the highly flammable soundproof foam on the ceiling, causing the blaze to spread rapidly. By around 1:18 AM, patrons became aware of the fire's danger and attempted to evacuate. However, the emergency exits on both the first and second floors were sealed, preventing an effective escape. This tragic incident resulted in the deaths of nine individuals and injuries to twelve others. At the time of the fire, approximately 200 people were inside the bar. When the ceiling caught fire, the audience mistakenly thought it was part of the fire performance. As a result, more and more pedestrians gathered at the fire site on the second floor, missing the best time to evacuate and increasing crowd

congestion, ultimately leading to significant casualties [39]. We also anonymized the crowd involved in the case, and the study does not include any personal sensitive information.

2.2. The coupling modeling

The paper simulates fire evacuation using the Fire Dynamics Simulator (FDS) combined with an Agent-Based Model (ABM). However, given the complexity of fire evacuation, as well as the limitations of model complexity and computational resources, several assumptions are made to simplify the model:

- (1) The impact of factors such as pedestrian panic, collisions, herd behavior, and helping behavior are not considered. Additionally, the crowd fully complies with instructions for staggered evacuation.
- (2) In terms of the environment, only factors such as obstacles, exits, and fire smoke are considered. It is assumed that exits are available, while other factors such as falling objects and ground damage are neglected.
- (3) Our model focuses on simulating the fire evacuation process and does not include external rescue efforts.

Fire Dynamics Simulator (FDS) is a powerful fire simulation software that can effectively calculate temperature and gas concentration changes in fire scenarios [42–43]. In FDS, the gas flow and heat transfer processes are calculated within the cells of the grid. In the ABM evacuation model, pedestrian movement also occurs within the patches (grid cells). Therefore, as shown in Fig. 2, data processing tools like Python can be used to process FDS simulation data, which can then be loaded into the NetLogo-based evacuation model within the same time steps as the crowd movement model. This allows for real-time influence of fire scenarios on pedestrian states within the cells, achieving dynamic coupling of fire scenarios and crowd movement behaviors. This method can capture the impact of fire hazards on pedestrians, effectively reproducing real fire crowd evacuation scenarios.

When handling three-dimensional disaster data from FDS in the two-dimensional space of the ABM model, certain factors must be considered. The following steps are typically taken to process the data:

1) The height of each Fire Dynamics Simulator (FDS) grid cell is equivalent to the floor height.
 2) Specifically, the time step for the model update frequency is set to 1 step/0.25 seconds, meaning that pedestrian states and disaster data are updated every 0.25s. Additionally, the grid cells for both the FDS and ABM models are set to 0.4m × 0.4m [44]. In this evacuation model, the state of each patch is either empty or occupied by pedestrians and obstacles. The boundaries of the model are composed of walls and exits. When pedestrians reach an exit, they are considered to have successfully evacuated. Furthermore, pedestrians' normal movement follows the standard Moore neighborhood, allowing movement in eight directions. Meanwhile, the movement speed of severely injured pedestrians is reduced, and deceased pedestrians lose the ability to move.

In this model, the calculation formula for the transfer probability P_{ij} of a normal pedestrian to an adjacent patch is as follows [37]:

$$P_{ij} = N_p^{-1} e^{k_s S_{ij} - k_D D_{ij} - k_F F_{ij} - k_R R_{ij}} (1 - n_{ij}) \quad (1)$$

$$N_p = \sum_i \sum_j P_{ij} \quad (2)$$

Where N_p is the normalization factor ensuring that $\sum P_{ij} = 1$. S_{ij} represents the value of a static field, with its sensitivity coefficient denoted by k_s . Each patch has a static field parameter S_{ij} , calculated based on the Dijkstra algorithm [45–46]. The Dijkstra algorithm computes the shortest path from each patch to the exit, generating a static potential field that accurately identifies the shortest path around obstacles, thereby ensuring the effectiveness and safety of path planning. The closer an area is to the exit, the larger the value of S_{ij} , indicating that the crowd will move toward areas nearer to the exit [47]. D_{ij} indicates the number of pedestrians at a given location, and k_D is the parameter for pedestrian crowding sensitivity. In Eq. (1), it is indicated that the crowd will try to move toward areas with fewer pedestrians to avoid congestion. The barrier repulsion field value is denoted by R_{ij} , with its sensitivity coefficient represented by k_R . It indicates that the crowd will try to move away from obstacles to enhance evacuation efficiency [48]. The repulsion field value of smoke to pedestrians is indicated by F_{ij} , and the sensitivity coefficient for the smoke repulsion field is denoted by k_F . During crowd movement, individuals will instinctively move towards areas with the lowest smoke concentration, with these values directly obtained from the Fire Dynamics Simulator (FDS) [42]. Following previous research [47–49] that considered similar conditions to those in our study, the fixed parameters were adjusted to $k_s = 1000$, $k_D = 2$, $k_R = 10$, and $k_F = 30$.

Additionally, the variable n_{ij} is used to represent the presence of obstacles within each patch grid cell (i, j). The following is the specific formula for the decision rule [50]:

$$n_{ij} = \begin{cases} 1, & \text{Occupied by walls or obstacles} \\ 0, & \text{Not occupied by walls or obstacles} \end{cases} \quad (3)$$

This indicates that the probability of the crowd moving toward walls or obstacles is 0.

Eqs. (1)–(3) indicate that pedestrians are more inclined to move toward directions that are closer to exits, have lower congestion levels, lower smoke concentration, and are farther from obstacles. Additionally, pedestrians cannot enter grids occupied by walls or obstacles.

2.3. The injury risk of pedestrians

In fire emergencies, high temperatures pose a severe threat to the health of evacuees. Purser highlighted that high temperatures in fires pose significant hazards to individuals attempting to escape, potentially resulting in heatstroke, skin burns, and respiratory tract burns. Exposure to temperatures exceeding 120 °C for just a few minutes can quickly

incapacitate individuals and may even lead to death [51]. Additionally, toxic gases produced by fires greatly threaten the health of evacuees, though it is challenging to fully assess the toxicity of each gas [52,53]. Therefore, this study focuses on analyzing the toxicity of carbon monoxide (CO). According to Haber's law, toxicity is dependent on the product of the cumulative dose, exposure time, and concentration [54]. Thus, there is a potentially fatal relationship between the concentration of inhaled CO (C_{CO}) and exposure time (t_{CO}). The specific formula is as follows (where k_1 is the correction coefficient, set to 0.8 in this case) [52]:

$$S_{CO} = k_1 \times C_{CO} \times t_{CO} \quad (4)$$

Experiments show that at carbon monoxide concentrations ranging from 1000 to 8000 ppm, primates lose consciousness when the cumulative exposure reaches 27,000 ppm·min. Therefore, this study sets the cumulative exposure of carbon monoxide to 27,000 ppm·min. Additionally, the "Handbook of Fire Protection Engineering" specifies the relationship between incapacitation time (t_{HEAT}) and temperature (T), as follows (where k_2 is the correction coefficient, set to 0.86 in this case) [54]:

$$t_{HEAT} = k_2 \times 5 \times 10^7 T^{-3.4} \quad (5)$$

Under disaster conditions, changes in the temperature field and carbon monoxide (CO) concentration will affect the environment of each pedestrian. Therefore, we use a cumulative damage model to calculate the health status of pedestrians at each time step. The specific formula is as follows:

$$H_p = 1 - \sum_v \left(\frac{C_{CO,v}}{S_{CO}} + \frac{1}{5 \times 10^7 T_v^{-3.4}} \right) \Delta t \quad (6)$$

In formula (6), $C_{CO,v}$ refers to the concentration of carbon monoxide at the pedestrian's location at time step v , while T_v is the temperature at the pedestrian's location at the same time step. The pedestrian's health index (H_p) initially starts at 1, and theoretically, H_p dropping to 0 indicates the pedestrian's death. However, in real fire scenarios, besides carbon monoxide and high temperatures, other harmful by-products may also be present. Therefore, this study sets 0.5 and 0.7 as thresholds for health assessment: pedestrians are considered seriously injured when H_p falls below 0.7, and considered dead when it falls below 0.5. Additionally, the movement speed of healthy pedestrians (V_f) is set to 1 m/s, and the movement speed of seriously injured pedestrians (V_h) is set to 0.4 m/s [53–54].

2.4. The stampede risk of pedestrians

Stampede accidents often occur in situations where large crowds gather [27]. To further study the risk of stampede accidents during fire evacuations, this study aims to determine the stampede risk value by calculating the pedestrian density near the exits. The formula is as follows [55]:

$$y_4 = \sum_v c g(\rho) \quad (7)$$

$$\rho_n = \frac{\sum_{(x,y) \in U} h(x,y)}{|U \times I^2|} \quad (8)$$

$$g(x) = \begin{cases} x, & \text{if } x \geq 4 \\ 0, & \text{if } x < 4 \end{cases} \quad (9)$$

$$h(x,y) = \begin{cases} 1, & \text{the cell}(x,y) \text{ is not empty} \\ 0, & \text{otherwise} \end{cases} \quad (10)$$

In formula (8), y_4 represents the overall stampede risk, v represents the time step, and c is the correction coefficient, set to 0.8 in this case [56]. ρ_n indicates the pedestrian density at the exit location. In formula

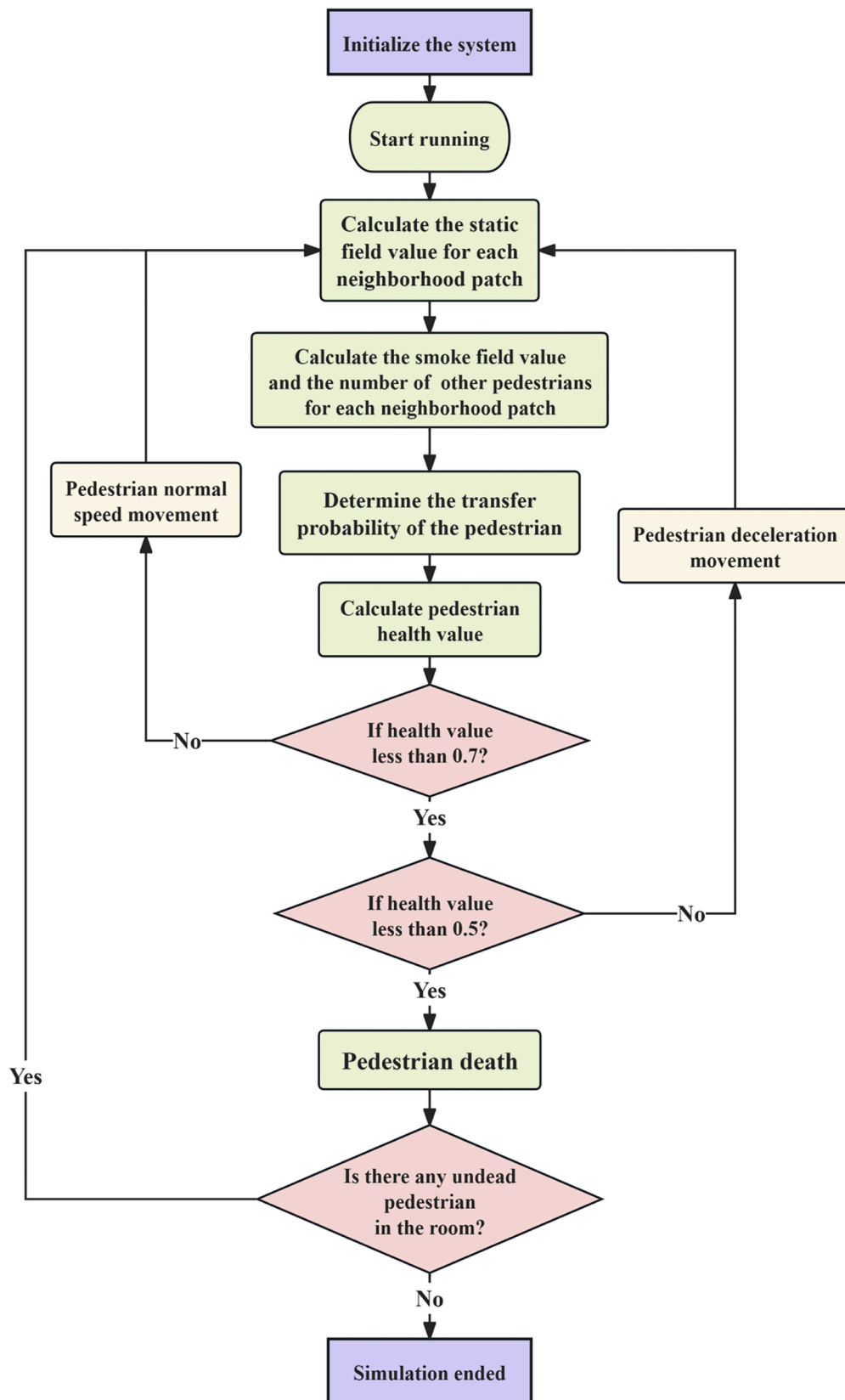


Fig. 3. Simulation update flowchart. Calculate the relevant field strength values to control crowd movement while updating the health values of pedestrians and their statuses.

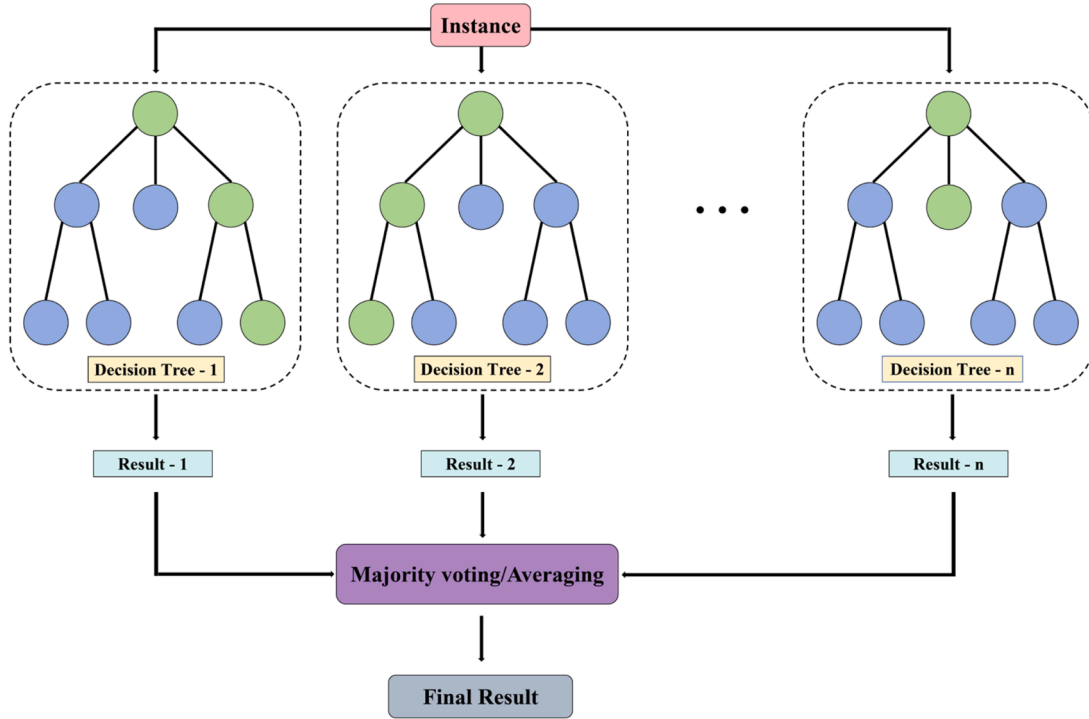


Fig. 4. Random Forest Model. Random Forest is an ensemble learning method that constructs multiple decision trees and combines their results for prediction and classification.

(8), U represents the area for stampede risk calculation. Since pedestrian congestion often occurs at the exits, this area is set near the first-floor exit. The variable l represents the side length of the block, set to 0.4 meters. Previous research has shown that stampede incidents occur when pedestrian density reaches 4 persons/m² [57]. We set the critical stampede risk density ρ_L to 4 persons/m². Therefore, when the pedestrian density ρ_n is less than 4 persons/m², the stampede risk is considered to be 0.

2.5. Strategy updating rules

The simulation first initializes the distribution of personnel, positioning the destination of pedestrians at the exits to facilitate evacuation. Next, the transfer probability of each pedestrian is calculated to determine their next position. After that, the overall stampede risk value is calculated, and each pedestrian's health value (H_p) and movement speed are updated. Pedestrians with H_p below the health threshold are considered dead. Pedestrians who reach the exit position evacuate the system safely in the subsequent time steps, indicating successful transfer. Thus, they no longer exist in the simulation or are included in further calculations. The simulation ends once all surviving pedestrians have successfully transferred. The specific update process is shown in Fig. 3.

2.6. Random forest prediction model

Optimizing crowd evacuation tasks presents the challenge of deciphering the intricate relationship between various influencing factors and evacuation objectives [58]. The advancement of computer-based simulations and machine learning algorithms has now made it feasible to address this complex task. Mainstream machine learning algorithms include linear regression, logistic regression, decision trees, neural networks, random forests, and support vector machines. Among them, Random Forest is an algorithm based on ensemble learning methods that improves model stability and accuracy through the combination of multiple decision trees. Compared to single models (such as a single decision tree, SVM, etc.), the predictive performance of Random Forest

is generally more reliable [59–60]. At the same time, Random Forest introduces randomness (such as Bootstrap sampling and random feature selection) to reduce the possibility of model overfitting. In some complex algorithms (such as neural networks), special tuning may be required to avoid overfitting [61]. Additionally, Random Forest can provide feature importance scores, which helps in understanding the basis of model decisions. In contrast, black-box models like neural networks often have weaker interpretability [62].

Random forests also exhibit a clear advantage in scalability, particularly when handling large-scale datasets. Since each tree is constructed independently, random forests can leverage parallel computing to accelerate the training process, making them highly effective for big data scenarios. In terms of computational complexity, the time complexity of random forests is proportional to the number and depth of trees. However, increasing the number of trees can enhance model accuracy without significantly adding computational overhead. Overall, random forests efficiently scale to large, high-dimensional datasets and offer strong parallel computing capabilities, making them exceptionally flexible and effective in practical applications [63]. Therefore, in this study, we will establish a Random Forest model to predict and analyze the outcomes of optimized crowd movement plans. The principles of the random forest model are illustrated in Fig. 4. In Section 4.2.2, we will apply the random forest model for data training and risk prediction.

2.7. PSO-GA optimization algorithm

The PSO-GA algorithm combines the characteristics of Particle Swarm Optimization (PSO) and Genetic Algorithm (GA), introducing the selection, crossover, and mutation operations of the genetic algorithm into the traditional particle swarm optimization algorithm to iteratively find the optimal solution. This hybrid algorithm can achieve better solutions when solving complex optimization problems, while overcoming the drawback of single algorithms easily falling into local optima. It improves the stability and global search capability of the algorithm [64].

The PSO-GA algorithm combines the strengths of Particle Swarm Optimization (PSO) and Genetic Algorithm (GA), enabling efficient

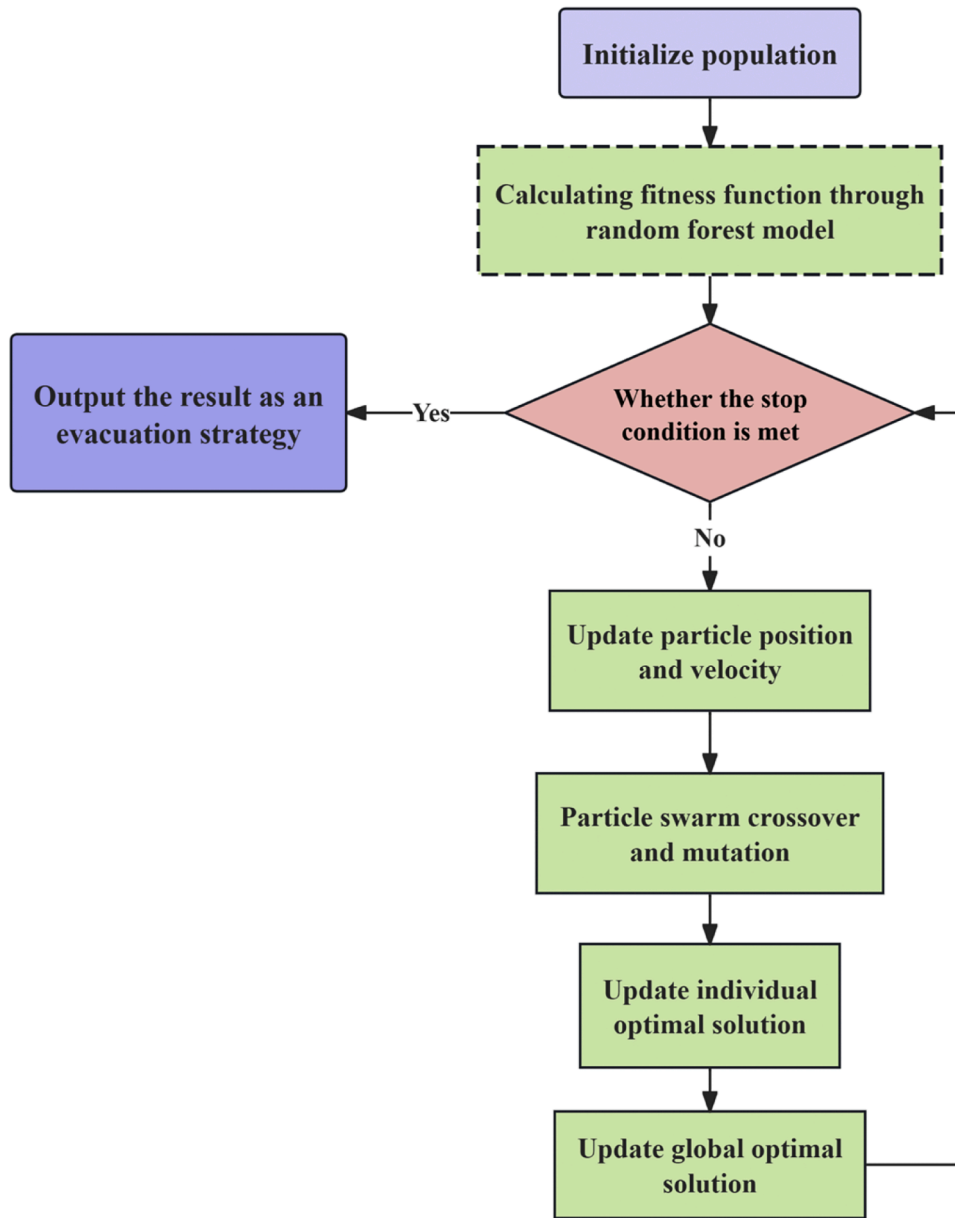


Fig. 5. PSO-GA algorithm optimization flowchart. This hybrid algorithm leverages the fast convergence of PSO and the global search capability of GA, aiming to improve optimization performance and the ability to solve complex problems.

handling of complex and large-scale optimization problems. With moderate computational complexity and independence from gradient information, it is well-suited for nonlinear and high-dimensional problems [65–66]. Both PSO and GA support parallel computing, offering strong scalability and accelerated optimization, particularly for large datasets. Additionally, PSO-GA demonstrates flexibility in addressing multi-objective and multi-constraint problems while maintaining robust performance, excelling in large-scale and high-dimensional scenarios [67–68]. In this study, we will use the PSO-GA algorithm to optimize evacuation plans, with the specific optimization process shown in Fig. 5.

In Section 4.2.3, we will apply the PSO-GA algorithm to further optimize and refine the evacuation strategy.

3. Real case simulation outcome

3.1. Scene construction

We modeled the simulated scenario by analyzing the internal

structure of the bar, as shown in Fig. 6. First, we established an FDS fire model with the fire source located at the ceiling of the second floor. The bar has two floors, each measuring $10\text{m} \times 8\text{m}$. Subsequently, the smoke data obtained from the FDS simulation were imported into the NetLogo software for crowd movement simulation, with 200 people randomly distributed within the bar.

3.2. Real fire case reconstruction and model accuracy verification

Based on the coupled model, we reproduced the process of the fire occurrence, as shown in Fig. 7A. When the fire broke out, the crowd did not evacuate promptly, and more pedestrians gathered on the second floor where the fire occurred. At this time, the health status of the pedestrians was good, and their color was blue. At time step $v=720$, the crowd realized the fire risk and began to evacuate urgently, with severely injured pedestrians appearing (colored yellow). At time step $v=1200$, the evacuation ended, with 9 pedestrians dying in the fire (red pedestrians represent deceased individuals). The 9 deceased pedestrians

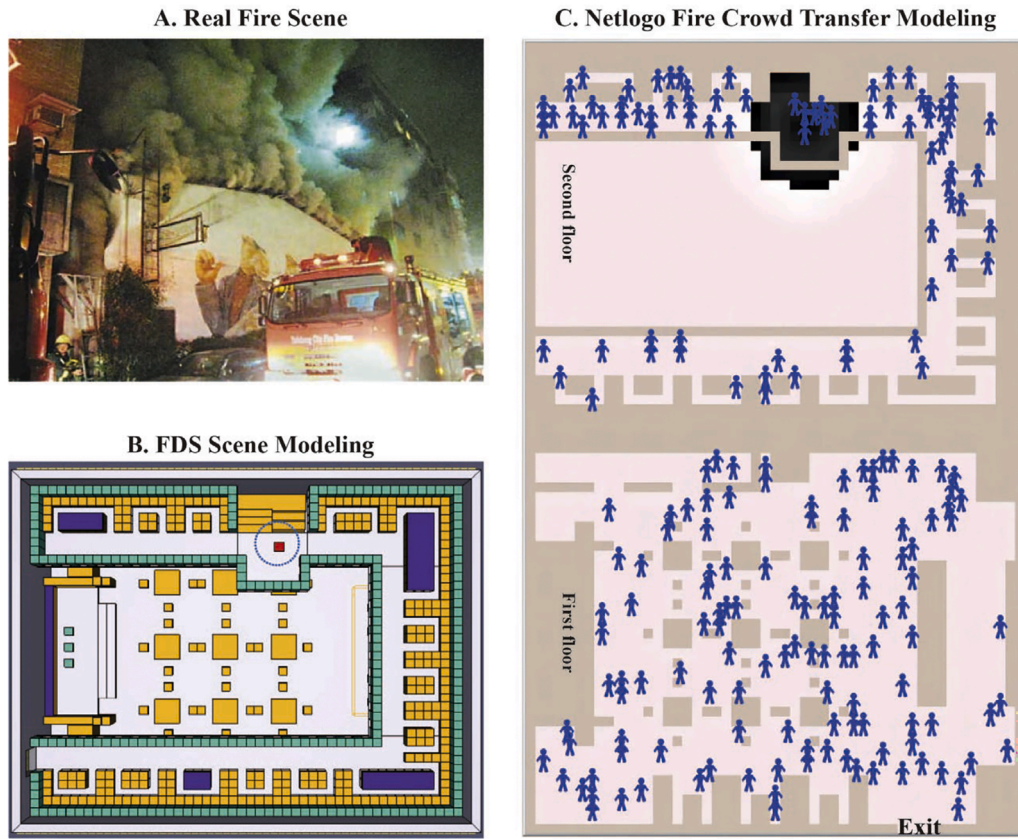


Fig. 6. Case Modeling. First, extract the relevant structural parameters of the building and perform FDS fire modeling based on the actual fire source location. Then, import the fire data into the NetLogo evacuation model to simulate the real fire process.

were located in the corridor near the second-floor stairway, which is consistent with the real situation.

The key indicators for evaluating the accuracy of this model are y_1 (total duration), y_2 (number of deaths), and y_3 (number of serious injuries). To ensure robust outcomes, we repeated the simulation 100 times and averaged the results. We used two criteria, effectiveness and robustness, to verify our optimal solution. Effectiveness requires that the simulation results closely match the real target case in terms of total duration (y_1), number of deaths (y_2), and number of serious injuries (y_3). Robustness indicates that this degree of fit should remain stable across repeated simulations [69].

(a) In Fig. 7B, the average value of the simulated total evacuation time (y_1) is 485.36, with an accuracy rate of 98.9% compared to the actual value (480). Meanwhile, the standard deviation (SD) is 12.01 ($N=100$), much lower than the mean. Most data points in the Q-Q plot are close to the 45° line, indicating that the simulation of the total evacuation time (y_1) is accurate and effective.

(b) In Fig. 7C, the average value of the simulated total number of deaths (y_2) is 9.05, with an accuracy rate of 99.45% compared to the actual value (9). Meanwhile, the standard deviation (SD) is 0.317 ($N=100$), significantly lower than the mean. Most data points in the Q-Q plot are close to the 45° diagonal line, indicating that the simulation results for the total number of deaths (y_2) are feasible.

(c) In Fig. 7D, the average value of the simulated total number of serious injuries (y_3) is 11.2, with an accuracy rate of 93.33% compared to the actual value (12). Meanwhile, the standard deviation (SD) is 0.451 ($N=100$), significantly lower than the mean. Most data points in the Q-Q plot are close to the 45° diagonal line, indicating that the simulation results for the total number of serious injuries (y_3) are also feasible.

Additionally, we conducted a Shapiro-Wilk test on the simulated data to assess its validity, with the detailed results presented in Table 1. It can be observed that the Statistic values for y_1 , y_2 , and y_3 all exceed 0.95, and the corresponding p-values are greater than 0.05, indicating that the data follows a normal distribution [70].

Fig. 7E shows the change in the number of evacuees over time, indicating a relatively smooth evacuation process. From Fig. 7F, it can be seen that the first deceased pedestrian appears at time step $v=831$. The number of deceased pedestrians significantly increases during the time steps $v=831$ to $v=917$, after which the number of deceased pedestrians no longer increases. This is because, after this period, all remaining pedestrians have left the second floor near the fire source, reducing the damage caused by the fire. In Fig. 7G, the first severely injured pedestrian appears at time step $v=784$. The number of severely injured pedestrians continues to rise for a period thereafter, with a slight decrease due to the death of some severely injured pedestrians, reaching a maximum of 13 at $v=909$. At $v=917$, the death of one severely injured pedestrian reduces the number of severely injured to 12, which remains stable for a period. At $v=1089$, the first severely injured pedestrian evacuates the room, and subsequently, all severely injured pedestrians evacuate, marking the end of the evacuation, with the number of severely injured pedestrians in the scene becoming 0.

4. Counterfactual experiments

Based on the verified accuracy and robustness of the model, we altered the scenario settings to obtain some counterfactual results, which can provide some guidance for future fire prevention and control efforts.

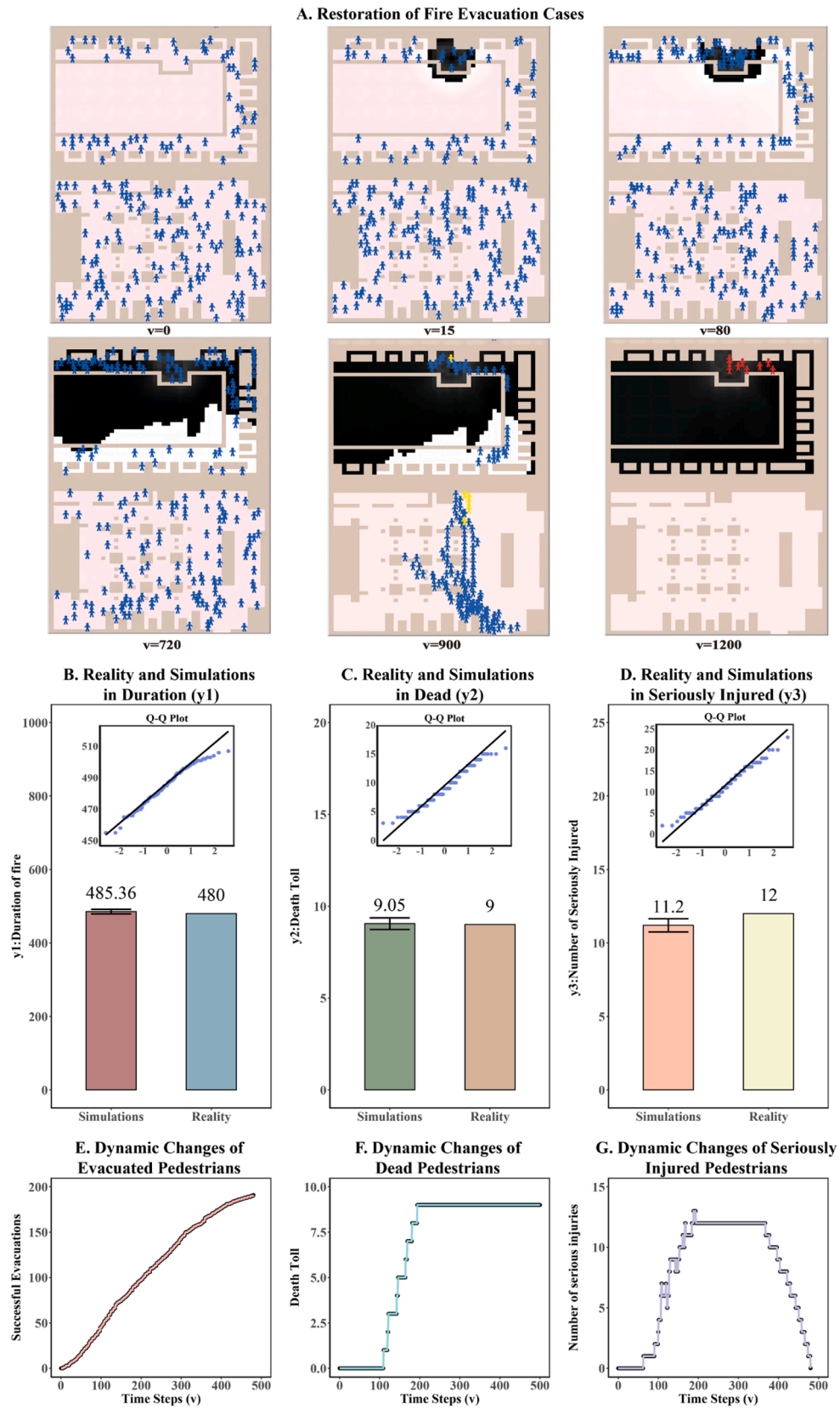


Fig. 7. Fire recreation process and numerical fitting. Fig. 7A shows the reproduction of the actual fire incident. Fig. 7B-D compare the average simulation values with the actual values. Fig. 7E-G provide a microscopic analysis of the evacuation process.

Table 1

Shapiro-Wilk test of simulation results.

Value Indicators	y1	y2	y3
Statistic	0.9753	0.9698	0.9831
p-value	0.05703	0.1611	0.2285

4.1. Impact of second floor exit

In this fire case, the efficiency of the evacuation was reduced, and the risk increased due to the closed escape exit on the second floor. Therefore, we simulated the scenario with the second-floor safety exit open and selected four risk evaluation indicators: total duration (y1), number of deaths (y2), number of serious injuries (y3), and stampede risk (y4). These were compared with the conditions in the real scenario where the safety exit was closed.

In Fig. 8B, it can be seen that when the second-floor exit is open, the average evacuation time (y1) is reduced by 22.47%, greatly enhancing evacuation efficiency. Fig. 8C shows that the average number of deceased pedestrians (y2) significantly decreases when the second-floor exit is open, almost approaching 0. However, the number of severely injured pedestrians increases by 28.8%. This is because the probability of death for severely injured pedestrians is greatly reduced, so the number of severely injured pedestrians increases in the end. The overall stampede risk value also decreases because the separation of evacuees through two exits significantly reduces the pedestrian density at the exits, as shown in Fig. 8G.

4.2. Social learning to enhance public fire safety

Social Learning Theory, introduced by psychologist Albert Bandura in the 1960s, underscores the importance of observation, imitation, and

modeling in learning. According to this theory, individuals acquire knowledge not only through their own direct experiences but also by observing the behaviors of others and the subsequent consequences [71]. In the face of sudden disasters, Social Learning Theory can play a crucial role in enhancing the coping abilities of communities and individuals, thereby improving overall societal safety. It enables individuals to make more informed decisions in emergency situations, such as fires, by computing and disseminating social knowledge. In the simulation, individuals adjust their behaviors based on the social knowledge they acquire, such as choosing the optimal evacuation route or deciding when to take action. This learning process leads to more effective collective behavior, thereby improving evacuation efficiency and reducing casualties. Thus, Social Learning Theory enhances group coordination and optimization during emergencies by guiding individual behavior [72].

In this fire case, people chose to watch the fire when it first started, unaware of its dangers, and missed the safe evacuation time. If public awareness of crisis and safety education is strengthened, allowing people to learn relevant social knowledge in advance and understand the dangers and severe consequences of fires, they would evacuate promptly instead of gathering to watch at the onset of a fire. This could significantly reduce casualties. By setting the evacuation response time T_D to

Table 2

Comparison of Two Evacuation Scenarios.

Evacuation scenario	Number of dead pedestrians	Number of seriously injured pedestrians	Average pedestrian health value
Realistic scenes	9.05	11.2	0.8874
Timely evacuation	0	0	0.9614

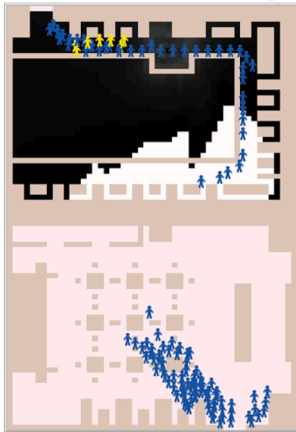
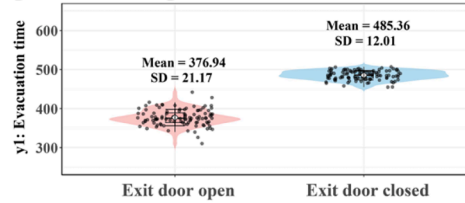
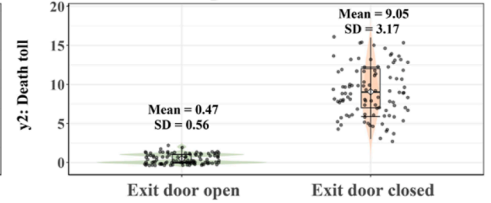
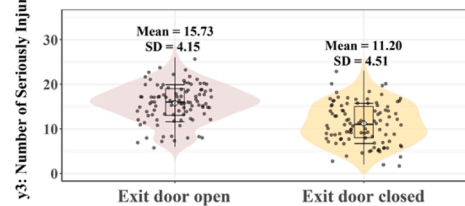
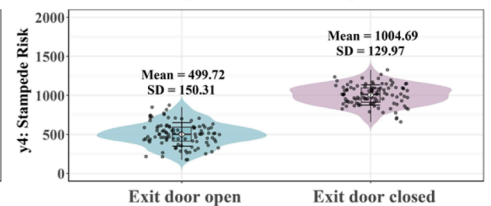
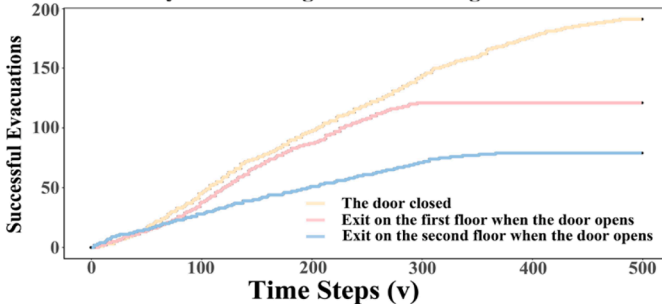
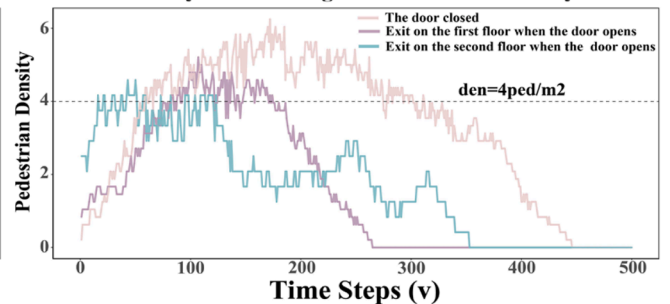
A. Scene of 2nd Floor Door Opening

B. Comparison of Evacuation Time

C. Comparison of Death Toll

D. Comparison of Seriously Injured

E. Comparison of Stampede Risk

F. Dynamic Changes of Evacuating Pedestrians

G. Dynamic Changes of Pedestrian Density


Fig. 8. Impact of the second-floor door closure on fire risk. Fig. 8A shows the evacuation process in the scenario where the second-floor exit is open. Fig. 8B-E compare various risk indicators with the real scenario. Fig. 8F-G provide a microscopic analysis of the evacuation dynamics.

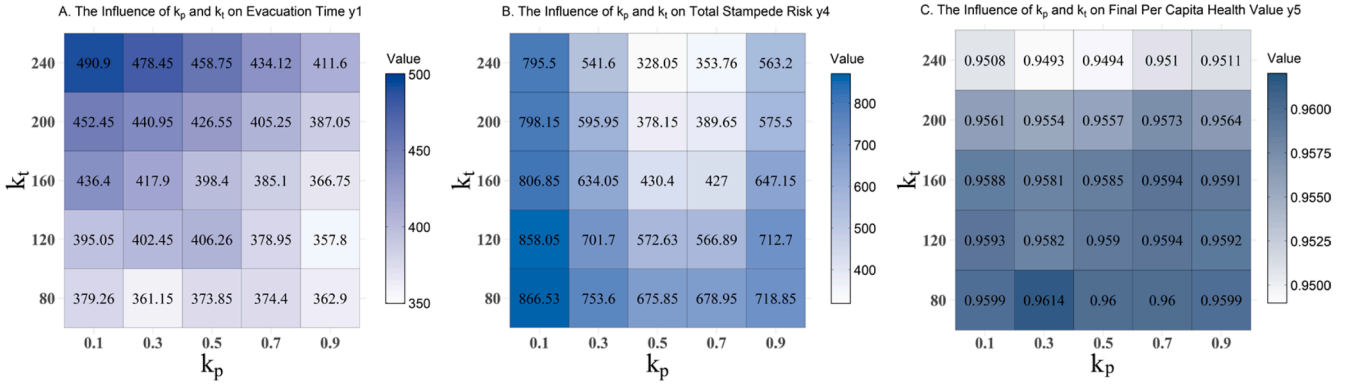


Fig. 9. Impact of k_p and k_t on evacuation outcomes (y_1 , y_4 , y_5). Figs. 9A, 9B, and 9C represent different risk indicators respectively.

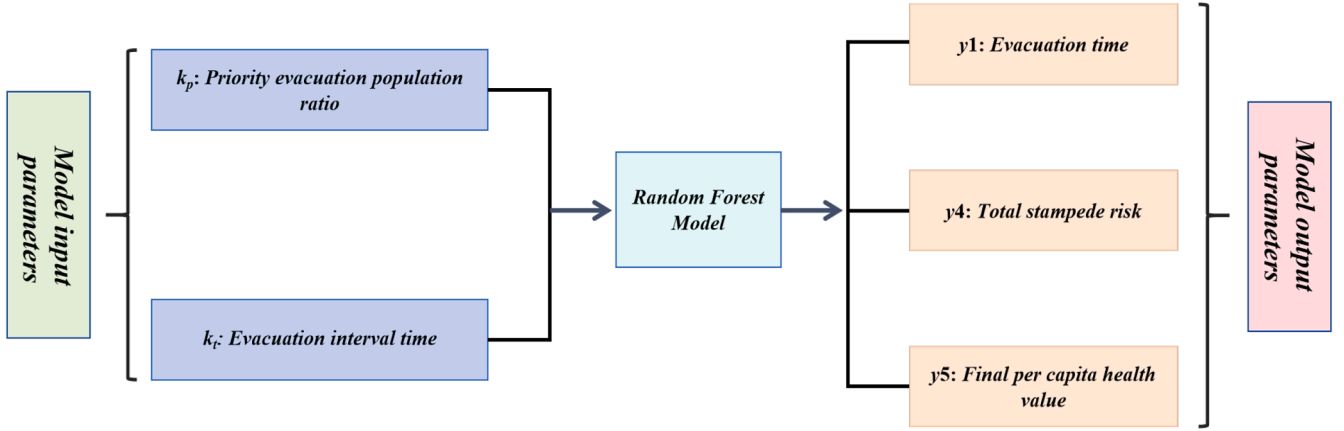


Fig. 10. Schematic diagram of the Random Forest model. In this model, there are two input parameters and three output parameters.

20 steps and avoiding gatherings in the second-floor fire area, as shown in Table 2, both the number of deceased and severely injured pedestrians dropped to 0, and the average health value of pedestrians increased by 8.34%.

4.2.1. Preliminary exploration of the role of phased evacuation

Through simulation methods, we acquired new social knowledge. We observed that when a large number of people rush to the exit simultaneously, it leads to congestion and increases the risk of serious stampedes. Implementing phased evacuation at the onset of a fire for large crowds can help reduce these risks and prevent fire-related casualties, thereby maximizing overall resident safety. Therefore, under the premise of timely evacuation of the second-floor crowd, we applied phased evacuation for the first-floor crowd. We set the proportion of timely evacuated people on the first floor to k_p and the waiting time for the second batch of evacuees to k_t . We selected total evacuation time (y_1), total stampede risk (y_4), and final average health value (y_5) as evaluation indicators to analyze the effects of different k_p (0.1, 0.3, 0.5, 0.7, 0.9) and k_t (80, 120, 160, 200, 240), as shown in Fig. 9.

In Fig. 9A, it can be seen that when $k_t \geq 160$, the evacuation time y_1 decreases as k_p increases, improving evacuation efficiency. This is because, at this point, the waiting time for the second batch of evacuees is too long, and the exit utilization efficiency is low when k_p is small. However, when $k_t \leq 120$, the change in y_1 with k_p is not significant, and the overall evacuation time decreases. The minimum value ($k_p=0.3$, $k_t=80$, $y_1=490.0$) is 35.91% lower compared to the maximum value ($k_p=0.1$, $k_t=240$, $y_1=361.15$).

From Fig. 9B, it can be seen that the stampede risk y_4 gradually increases as k_t decreases; the longer the evacuation interval, the lower the stampede risk. However, y_4 initially decreases and then increases with

the increase of k_p , with the minimum stampede risk value at $k_p=0.5$ because the total density of people at the exit is the lowest at this point.

The values of k_p and k_t have little impact on the final average health value y_5 , as shown in Fig. 9C. This is because the delayed evacuees are all located on the first floor, while the high temperatures and toxic gases during the fire are mainly on the second floor, having minimal impact on the delayed evacuees on the first floor. However, in general, an increase in k_t leads to a decrease in the average health value of pedestrians. Regardless of the values of k_p and k_t , the number of severely injured and deceased pedestrians remains 0.

4.2.2. Random forest prediction of fire risk

Here, we use a Random Forest model to predict the strategies for phased evacuation. The input and output parameters of the model are shown in Fig. 10, where the input parameters are the proportion of the priority evacuation crowd k_p and the evacuation interval time k_t , while the output parameters are y_1 , y_4 , and y_5 . By selecting more varied values of k_p (range: 0–1) and k_t (range: 60–280) as inputs to obtain data, we train this model with a large dataset to further explore the impact of k_p and k_t on fire risk.

After filtering out outliers in the simulation data, we selected 2000 sets of data. Among them, 1600 sets of data were used as the training set, and 400 sets of data were used as the test set. R^2 (R-squared) and MSE (Mean Squared Error) were used as evaluation metrics for model accuracy [73]. Additionally, during the construction of each decision tree, the Random Forest records the impurity reduction brought by each feature at each split node. The average impurity reduction value of each feature across all trees is calculated as the importance indicator of that feature [74].

Moreover, we integrated SHAP (Shapley Additive Explanation) into

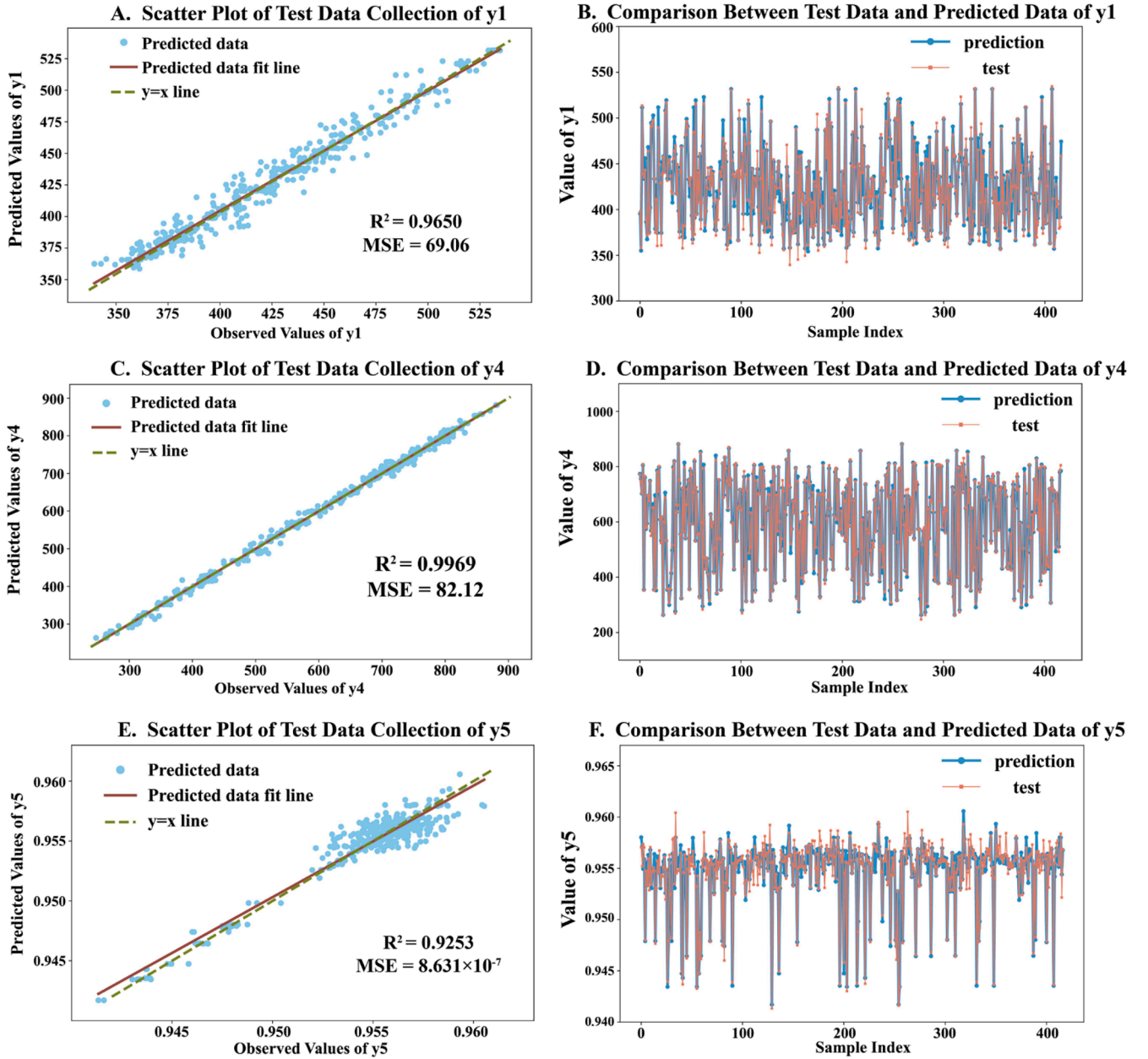


Fig. 11. Random Forest model accuracy testing. The left graph shows the fitting analysis of the test data, and the right graph compares the test values with the predicted values.

the proposed method to explain and analyze the importance of influencing factors in the established Random Forest model. SHAP uses a game theory-based Shapley value method to provide transparent, fair, and model-agnostic explanations. It quantifies the contribution of each feature to the model's predictions, thereby enhancing the interpretability and credibility of the model. SHAP is suitable for explaining both global and local predictions [75]. This interpretability enhances the model's transparency and reliability, enabling emergency management personnel to better understand and trust the factors driving the model's predictions. Consequently, they can develop more effective emergency response and prevention strategies. SHAP is applicable to both global and local prediction explanations, providing robust support for fire risk assessment and decision-making.

The results regarding model accuracy and feature importance are shown in Figs. 11 and 12.

(a) From Fig. 11A and Fig. 11B, it can be seen that the prediction of y_1 has $R^2=0.9650$, and MSE (69.06) is much smaller than the mean. The fitted line of the predicted data is close to the 45° line, indicating high correlation and accuracy. Additionally, Fig. 12A shows that the main factor influencing evacuation time is the interval time k_t , with an influence index of 0.763.

(b) As shown in Fig. 11C and 11D, the model's prediction of y_4 as an output has $R^2=0.9969$, with the fitted line of the predicted data almost coinciding with the 45° line. MSE (82.12) is also much smaller than the mean, indicating that the model's prediction is highly reliable. Furthermore, Fig. 12B shows that the influence index of k_p (0.950) is slightly higher than that of k_t (0.410). This is because the ratio of the two batches of evacuated people more significantly affects the congestion level at the exit.

(c) In Fig. 11E and 11F, the prediction of y_5 has $R^2=0.9253$, with the fitted line of the predicted data also closely approximating the 45° line. MSE (8.631×10^{-7}) is within the normal range, indicating that

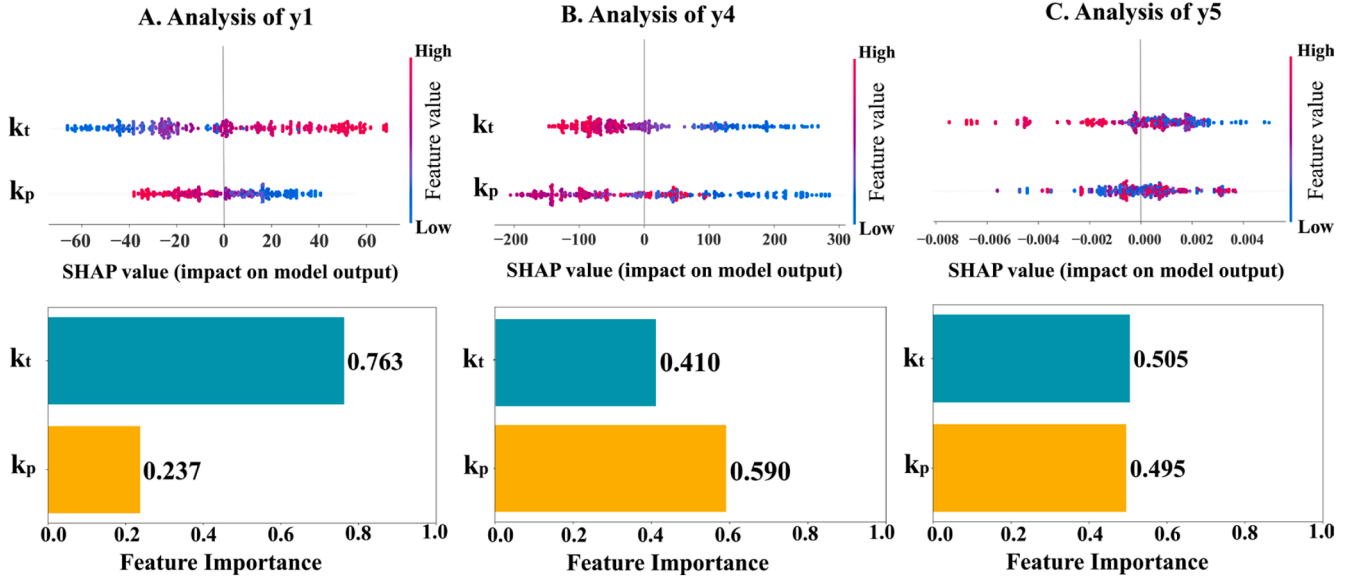


Fig. 12. The Importance of Random Forest Features and SHAP Analysis. The upper part shows the SHAP analysis, while the lower part describes feature importance. This figure primarily demonstrates the extent to which different input parameters influence the results.

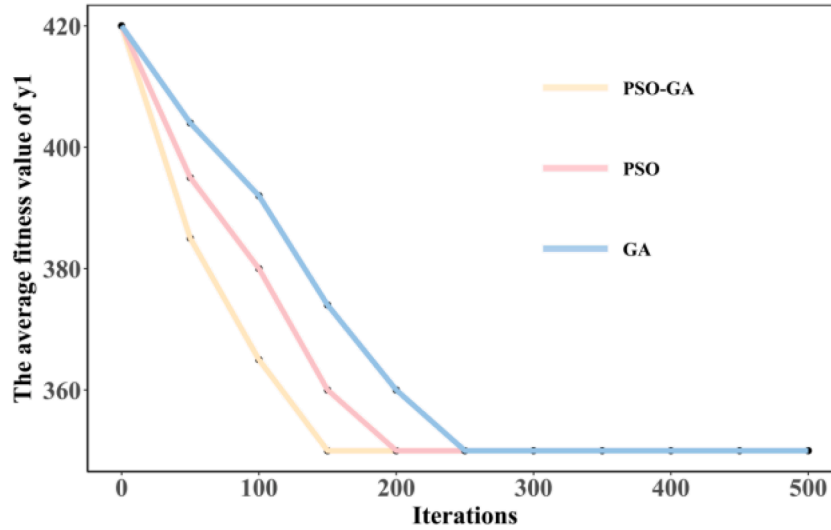


Fig. 13. Comparison of Optimization Algorithms. The optimization speeds of the PSO-GA, PSO, and GA algorithms are compared, demonstrating the effectiveness of the PSO-GA algorithm.

the model's prediction is also very accurate. Additionally, it can be seen that the predicted values of y_5 are mainly concentrated in the range [0.9526, 0.9598]. Fig. 12C shows that the influence of k_p (0.505) and k_t (0.495) on the results is nearly equal.

In summary, the Random Forest model demonstrates good reliability and accuracy in predicting the data. Testing the predicted values for y_1 , y_4 , and y_5 shows that R^2 values are all greater than 0.9, and the MSE values are much lower than the mean of the samples. Additionally, the model provides feature importance scores, helping us understand which features are most important in the model and offering deeper insights into the model's decision-making process.

4.2.3. Particle swarm optimization-genetic algorithm (PSO-GA) for optimizing evacuation strategies

We used the Random Forest model to predict fire risks (y_1 , y_4 , y_5) under different k_t and k_p conditions. Here, by applying the PSO-GA algorithm combined with the Random Forest model, we established a

multi-objective optimization model to test the impact of different k_t and k_p on fire risks and select the optimal evacuation plans. Therefore, we defined three optimization objectives, with the specific formulas as follows:

$$\begin{cases} \min y_1 \\ \min y_4 \\ \max y_5 \end{cases} \quad (11)$$

From Eq. (11), it is evident that when evaluating the safety of an evacuation plan, we aim to minimize the evacuation time (y_1), minimize the risk of stampede (y_4), and maximize the average life value (y_5). Additionally, we establish two constraint conditions to control the ranges of the input values k_t and k_p , as specified in formulas (12):

$$\begin{cases} 40 \leq k_p \leq 280 \\ 0 < k_t < 1 \end{cases} \quad (12)$$

In this optimization algorithm, we set the inertia weight $w_1=0.6$,

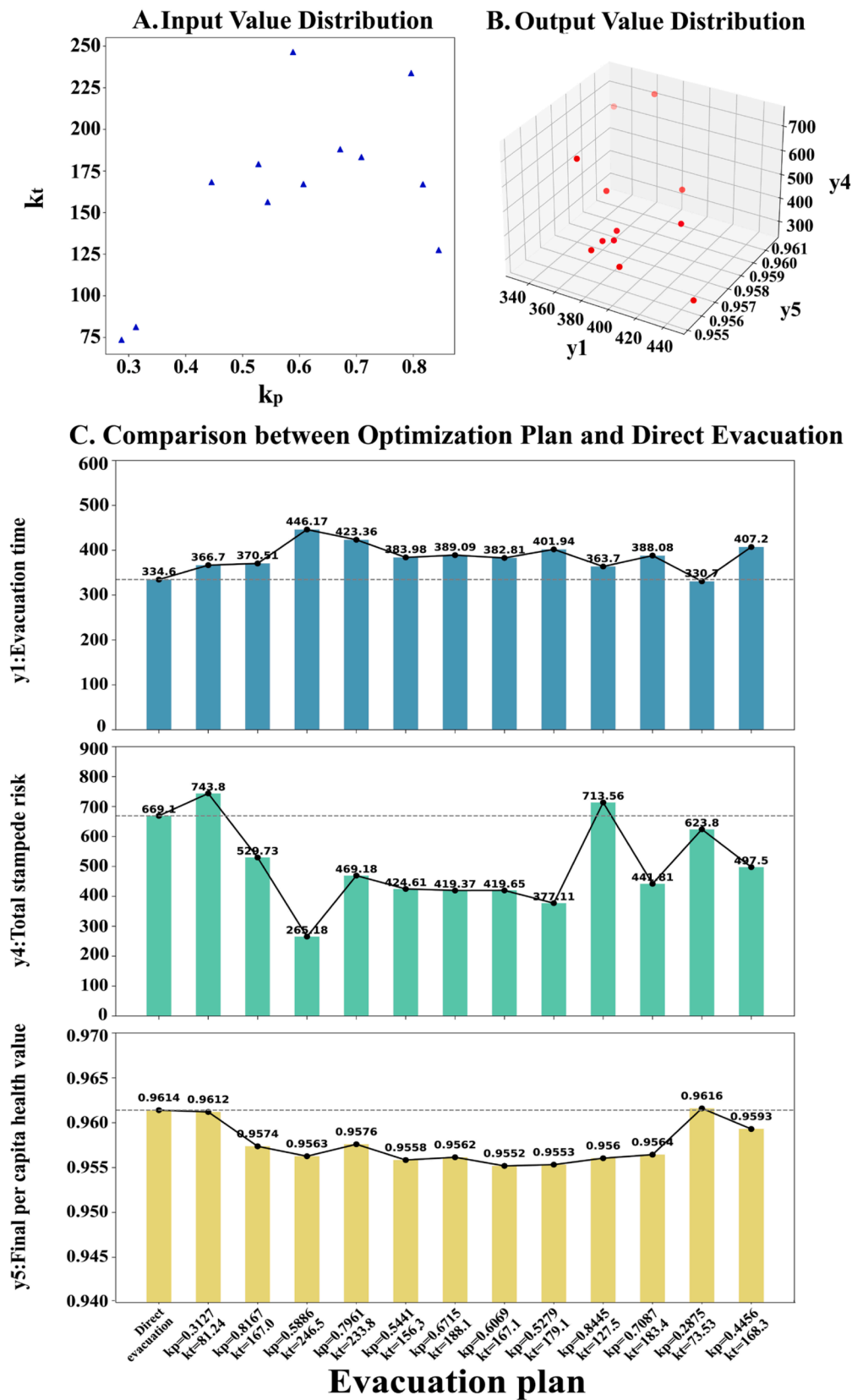


Fig. 14. Distribution of Optimized Data and Comparison of Schemes. Fig. 14A and Fig. 14B show the distribution of the input and output data of the optimization results, respectively. Fig. 13C compares the three risk indicators of the optimized schemes with the results of direct evacuation.

cognitive constant $c_1=1.6$, social constant $c_2=1.6$, crossover rate $Cr=0.8$, mutation rate $Mr=0.2$, initial population size $n_p=50$, and maximum iterations $M_i=1000$ [76,77].

In Fig 13, we compare the optimization processes of three algorithms. It can be observed that the PSO-GA algorithm converges the fastest and remains stable. This indicates that the convergence speed of the PSO-GA algorithm is significantly superior to that of the individual PSO or GA algorithms.

The optimization results are shown in Fig. 14, where a total of 12 Pareto frontier solutions were obtained. Fig. 14A shows the distribution of optimized input values, while Fig. 14B shows the distribution of optimized output values. From Fig. 14, it can be observed that 9 out of the 12 optimized schemes have partial target values that are superior to the direct evacuation scenario. These schemes generally reduce the risk of stampede but simultaneously increase evacuation time and the danger level to pedestrian life.

Moreover, when $k_p=0.3127$, $k_t=81.24$ and $k_p=0.8445$, $k_t=127.5$, the risk of stampede is higher than that of direct evacuation. This is because, in these two scenarios, a large number of evacuees on the first floor did not evacuate in time, while the crowd on the second floor surged towards the exit, further exacerbating the risk of stampede and reducing evacuation efficiency. Conversely, when $k_p=0.2875$ and $k_t=73.53$, all three target values are superior to direct evacuation. Specifically, y_1 decreased by 1.17%, y_4 decreased by 6.77%, and y_5 (average life value of pedestrians) increased by 0.208%. This is because, under this scheme, the exit congestion is reduced while maximizing the evacuation efficiency of the exit, thereby mitigating the harm caused by the fire to the human body.

Therefore, the evacuation scheme with $k_p=0.2875$ and $k_t=73.53$ is considered the optimal scheme, while the 9 schemes with some target values superior to direct evacuation are considered suboptimal schemes. The two schemes with no advantageous target values are deemed ineffective. By acquiring relevant social knowledge, individuals learn to wait for a certain period during emergencies, thereby avoiding a rush to the exit. This adjustment in collective behavior not only reduces the risk of congestion and stampedes but also maximizes safety throughout the entire evacuation process.

However, implementing phased evacuation strategies in real situations presents certain challenges. In actual evacuations, individuals may not follow instructions due to panic. The public's understanding and acceptance of phased evacuation strategies may be limited, especially in the absence of regular drills. Therefore, it is recommended to increase awareness of phased evacuation strategies and conduct regular evacuation drills to improve the public's understanding and ability to execute these strategies. Additionally, specialized training should be provided to building managers and safety personnel to enhance their capacity to organize phased evacuations in emergency situations.

5. Conclusion

Fire accidents typically lead to a range of social issues, exacerbating societal crises. These issues include displacement and homelessness, economic hardship, mental health problems, community disintegration, public health hazards, and educational disruption. The destruction caused by fires not only affects the quality of life for individuals and families but also poses long-term adverse impacts on the social structure and economic stability of entire communities. Therefore, it is crucial to explore the mechanisms of crowd behavior during fire incidents, optimize evacuation strategies, and promote relevant social safety knowledge. Based on this, our study proposes a fire evacuation model that couples the Fire Dynamics Simulator (FDS) and Agent-Based Modeling (ABM) to reproduce a bar fire incident that occurred in Taiwan in 2011. This model incorporates the concept of social learning and evaluates risk using various indicators such as evacuation time, stampede risk, casualties, and pedestrian health status. By applying machine learning combined with intelligent optimization methods, we aimed to optimize

evacuation strategies. We have identified several conclusions that can guide future research:

- (1) The coupling of the FDS and ABM models for fire simulation can fully leverage the advantages of both, achieving comprehensive fire simulations from fire physical dynamics to human behavior response. This not only helps improve the effectiveness of fire emergency response but also optimizes evacuation strategies, designs safer building environments, and enhances the fire response capabilities of the public and emergency personnel.
- (2) In this case, the safety exit on the second floor was closed when the fire occurred, which significantly reduced the efficiency of escape. In our counterfactual simulation, setting the second-floor escape route to open greatly ensured crowd safety and improved evacuation efficiency. Therefore, maintaining the accessibility of all safety exits is crucial.
- (3) Social knowledge plays a crucial role in enhancing public safety during a fire incident. In this case, the crowd chose to watch the fire rather than evacuate promptly, missing the optimal evacuation window. If the crowd had prior knowledge of relevant social protocols or if on-site staff had organized timely evacuations, public safety could have been significantly ensured. This study shows that if the crowd chooses to evacuate at the onset of the fire, the risk of casualties can be reduced to zero.
- (4) Appropriate phased evacuation strategies can enhance the efficiency of exit usage, reducing evacuation time and minimizing the risk of stampedes. In this study, the method combining random forest with the PSO-GA algorithm effectively identifies optimal evacuation strategies. This approach can generate relevant social knowledge, offering guidance and insights for developing emergency evacuation strategies in various public settings.
- (5) Limitations and Future Work. Although our developed model is highly idealized, it provides valuable perspectives for addressing complex real-world fire scenarios. This study primarily focuses on the impact of fire environments on pedestrian physical health, without considering the effects of factors such as smoke on pedestrians' psychological state and visibility. And the influence of environmental factors and external rescue on the evacuation model is not further considered. Future research should delve deeper into how these fire-related factors affect pedestrians' psychological state and visual range, as well as the dynamic interactions between pedestrians.

At the same time, more complex environmental factors and emergency rescue measures should be further considered to more accurately replicate the real fire evacuation process. Moreover, when applying this model to the development of real-world evacuation strategies, the impact of environmental conditions and emergency measures must be fully considered. The primary focus should be on fire prevention design in buildings, with evacuation strategy development as a secondary concern. At the same time, the simulation results of this model should be further validated through real evacuation experiments to mitigate the risk of misuse of the simulated data. Our research holds significant importance and social value for public safety. Through in-depth analysis, potential fire hazards can be more effectively predicted and prevented, thereby enhancing public safety.

CRediT authorship contribution statement

Peng Lu: Writing – original draft, Resources, Funding acquisition, Conceptualization. **Yufei Li:** Writing – review & editing, Writing – original draft, Supervision, Software, Investigation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the National Social Science Foundation of China (Grant No. 18VXK005, 19ZDA143 & 21ASH003), Beijing Social Science Foundation (Grant No. 16SRB014), and the Fundamental Research Funds for the Central Universities of Central South University (Grant No. 2023ZZTS0526).

Data availability

The data that has been used is confidential.

References

- [1] Zheng Y, Bin Jia, Li XG, et al. Evacuation dynamics considering pedestrians' movement behavior change with fire and smoke spreading[J]. *Safety Sci* 2016;92: 180–9.
- [2] Haase K, Kasper M, Koch M, et al. A pilgrim scheduling approach to increase safety during the Hajj[J]. *Oper Res* 2019;67:376–406.
- [3] Burkle FM, Hsu EB. Ram Janki Temple: understanding human stampedes[J]. *Lancet* 2011;377:106–7.
- [4] Zhu RX, Lin D, Jendryke M, et al. Geo-tagged social Media data-based analytical approach for perceiving impacts of social events[J]. *ISPRS Int J Geo-Inf* 2019;8:15.
- [5] Xin J, Huang C. Fire risk analysis of residential buildings based on scenario clusters and its application in fire risk management[J]. *Fire Safety J* 2013;62(4):72–8.
- [6] Chuvieco E, Yebra M, Martino S, et al. Towards an integrated approach to wildfire risk assessment: when, where, what and how may the landscapes burn[J]. *Fire* 2023;6(5):215.
- [7] Hu J, Xucai X, Shu X, et al. Fire risk assessments of informal settlements based on Fire risk Index and Bayesian network[J]. *Int J Environ Res Public Health* 2022;19: 15689.
- [8] Landucci G, Argenti F, Tugnoli A, et al. Quantitative assessment of safety barrier performance in the prevention of domino scenarios triggered by fire[J]. *Reliab Eng Syst Safe* 2015;143:30–43.
- [9] Li X, Yu H, Song W, et al. Fundamental diagrams for straight movement of pedestrians in water[J]. *Safety Sci* 2023;159:106018.
- [10] Seike M, Kawabata N, Hasegawa M. Experiments of evacuation speed in smoke-filled tunnel[J]. *Tunnelling and Underground Space Technology* 2016;53:61–7.
- [11] Mossberg A, Nilsson D, Andr  e K. Unannounced evacuation experiment in a high-rise hotel building with evacuation elevators: a study of evacuation behaviour using eye-tracking[J]. *Fire Technol* 2021;57:1259–81.
- [12] Cao S, Wang P, Yao M, et al. Dynamic analysis of pedestrian movement in single-file experiment under limited visibility[J]. *Commun Nonlinear Sci Numer Simul* 2019;69:329–42.
- [13] Jiang A, Mo Y, Kalasapudi VS. Status quo and challenges and future development of fire emergency evacuation research and application in built environment[J]. *J Inf Technol Constr* 2022;27:781–801.
- [14] Zhang Y, Huang X. A review of tunnel fire evacuation strategies and state-of-the-art research in China[J]. *Fire Technol* 2022;1–34.
- [15] Hostetter H, Naser MZ. Characterizing disability in fire evacuation: a progressive review[J]. *J Build Eng* 2022;53:104573.
- [16] Zikeloglou I, Lekkas E, Lozios S, et al. Community's evacuation planning and response for the 2021–2022 fire seasons in Greece[J]. *Safety Sci* 2024;172:106434.
- [17] Kuo TW, Lin CY, Chuang YJ, et al. Using smartphones for indoor fire evacuation[J]. *Int J Environ Res Public Health* 2022;19:15689.
- [18] Liu Z, Li Y, Zhang Z, et al. A new evacuation accessibility analysis approach based on spatial information[J]. *Reliab Eng Syst Safe* 2022;222:108395.
- [19] Xu Z, Wei W, Wei J, et al. Virtual drill for indoor fire evacuations considering occupant physical collisions[J]. *Autom Constr* 2020;109:102999.
- [20] Dulebenets M A, Abioye O F, Ozguven E E, et al. Development of statistical models for improving efficiency of emergency evacuation in areas with vulnerable population[J]. *Reliab Eng Syst Safe* 2019;182:233–49.
- [21] Wang Y, Ma W, Wang T, et al. Dynamic optimisation of evacuation route in the fire scenarios of offshore drilling platforms[J]. *Ocean Eng* 2022;247:110564.
- [22] Li XJ, Chen WB, Chen RX, et al. Evacuation-path-selection model of real-time fire diffusion in urban underground complexes[J]. *Comput Ind Eng* 2023;177:109014.
- [23] Khakzad N. A methodology based on Dijkstra's algorithm and mathematical programming for optimal evacuation in process plants in the event of major tank fires[J]. *Reliab Eng Syst Safe* 2023;236:109291.
- [24] Junfeng C, Maohua Z, Peiyun Q, et al. Map the fire risk in buildings: A hybrid method of ASET-RSET concept and FED concept[J]. *Reliab Eng Syst Safe* 2023;240: 109571.
- [25] Fang ZM. Modelling and experimental study of evacuation process considering the effect of fire[D]. *Univ Sci Technol China*; 2012.
- [26] Ding Z, Xu S, Xie XF, et al. A building information modeling-based fire emergency evacuation simulation system for large infrastructures[J]. *Reliab Eng Syst Safe* 2024;244:109917.
- [27] Helbing D, Farkas I, Vicsek T. Social force model for pedestrian dynamics[J]. *Nature* 2000;407:487–90.
- [28] Liu Q, He R, Zhang L. Simulation-based multi-objective optimization for enhanced safety of fire emergency response in metro stations[J]. *Reliab Eng Syst Safe* 2022; 228:108820.
- [29] HH Yen, Lin CH, Tsao HW. Novel smoke-aware individual evacuation and congestion-aware group evacuation algorithms in IoT-enabled multi-story multi-exit buildings[J]. *IEEE Access* 2022;10:119402–18.
- [30] Chi JH. Using FDS program and an evacuation test to develop hotel fire safety strategy[J]. *J Chin Inst Eng* 2014;37:288–99.
- [31] Lee J, Lee M, Jun C. Fire evacuation simulation considering the movement of pedestrian according to fire spread[C]. In: *Int Conf Geomat Geosp Technol*; 2018. p. 273–81.
- [32] Li Z, Huang H, Li N, et al. An agent-based simulator for indoor crowd evacuation considering fire impacts[J]. *Autom Constr* 2020;120:103395.
- [33] Lu P, Zhang Z, Onyebuchi CH, et al. Agent-based modeling of high-rise building fires reveals self-rescue behaviors and better fire protection designs[J]. In: 127; 2024, 107401.
- [34] Kim G, Heo G. Agent-based radiological emergency evacuation simulation modeling considering mitigation infrastructures[J]. *Reliab Eng Syst Safe* 2023;233: 109098.
- [35] Lu P, Li Y, Wen F, et al. Agent-based modeling of mass shooting case with the counterforce of policemen[J]. *Complex Intell Syst* 2023;9(5):5093–113.
- [36] Sun L, d'Ayala D, Fayjaloun R, et al. Agent-based model on resilience-oriented rapid responses of road networks under seismic hazard[J]. *Reliab Eng Syst Safe* 2021;216:108030.
- [37] Guo C, Huo F, Li Y, et al. An evacuation model considering pedestrian crowding and stampede under terrorist attacks[J]. *Reliab Eng Syst Safe* 2024;249:110230.
- [38] Liu L, Zhang H, Shi J, et al. Real-time evacuation route optimization in the fire scenarios of cruise ships[J]. *Simul Model Pract Theory* 2023;129:102843.
- [39] Lin CS, Chen TC, Lee TY. Fire computer-simulated fire scene verification of entertainment establishments[J]. *Appl Mech Mater* 2013;372:630–4.
- [40] Lu P, Li Y, Wen F, et al. Social knowledge enhances collective safety: computational models and simulations[J]. *IEEE Trans Comput Soc Syst* 2022;10(2):807–18.
- [41] Wang H, Cai W, Wang Y. Optimization of a hybrid ejector air conditioning system with PSOGA[J]. *Appl Therm Eng* 2017;112:1474–86.
- [42] Wang X, Liu Z, Loughney S, et al. Numerical analysis and staircase layout optimisation for a Ro-Ro passenger ship during emergency evacuation[J]. *Reliability Engineering & System Safety* 2022;217:108056.
- [43] Sakurahara T, Mohaghegh Z, Reihani S, et al. An integrated methodology for spatio-temporal incorporation of underlying failure mechanisms into fire probabilistic risk assessment of nuclear power plants[J]. *Reliability Engineering & System Safety* 2018;169:242–57.
- [44] Cao SC, Song WG, Liu XD, et al. Simulation of pedestrian evacuation in a room under fire emergency[J]. *Procedia Eng* 2014;71:403–9.
- [45] Zyczkowski M, Szlapczynski R. Collision risk-informed weather routing for sailboats[J]. *Reliability Engineering & System Safety* 2023;232:109015.
- [46] Mirahadi F, McCabe BY. EvacuaSafe: A real-time model for building evacuation based on Dijkstra's algorithm[J]. *J Build Eng* 2021;34:101687.
- [47] Zheng Y, Jia B, Li XG, et al. Evacuation dynamics considering pedestrians' movement behavior change with fire and smoke spreading[J]. *Safety Sci* 2016;92: 180–9.
- [48] Song C, Shao Q, Zhu P, et al. An emergency aircraft evacuation simulation considering passenger overtaking and luggage retrieval[J]. *Reliab Eng Syst Safe* 2023;229:108851.
- [49] Chen CK, Li J, Zhang D. Study on evacuation behaviors at a T-shaped intersection by a force-driving cellular automata model[J]. *Physica A* 2012;391:2408–20.
- [50] Zheng Y, Li XG, Jia B, et al. Simulation of pedestrians' evacuation dynamics with underground flood spreading based on cellular automaton[J]. *Simul Model Pract Theory* 2019;94:149–61.
- [51] Purser DA, McAllister JL. Assessment of hazards to occupants from smoke, toxic gases, and heat, *sfpe handbook of fire protection engineering*, 5th ed. New York: Springer; 2016. p. 2308–428 [J].
- [52] Levin BC. New research avenues in toxicology: 7-gas N-gas model, toxicant suppressants, and genetic toxicology[J]. *Toxicology* 1996;115:89–106.
- [53] Li Z, Huang H, Li N, Chu ML, Law K. An agent-based simulator for indoor crowd evacuation considering fire impacts[J]. *Autom Constr* 2020;120:103395.
- [54] Fang Z, Lv W, Li X, Song W. A multi-grid evacuation model considering the threat of fire to human life and its application to building fire risk assessment[J]. *Fire Technol* 2019;55:2005–26.
- [55] Huo FZ, Li YF, Li C, et al. An extended model describing pedestrian evacuation considering pedestrian crowding and stampede behavior[J]. *Physica A* 2022;604: 127907.
- [56] Yi JX, Pan SL, Chen Q. Simulation of pedestrian evacuation in stampedes based on a cellular automaton model[J]. *Simul Model Pract Theory* 2020;104:102147.
- [57] Ibrahim AM, Venkat I, Subramanian KG. Intelligent Evacuation Management Systems: A review[J]. *ACM Trans Intell Syst Technol* 2016;7.
- [58] Lu P, Wen F, Li Y, et al. Multi-agent modeling of crowd dynamics under mass shooting cases[J]. *Chaos Solitons Fractals* 2021;153:111513.
- [59] Rhodes JS, Cutler A, Moon KR. Geometry-and accuracy-preserving random forest proximities[J]. *IEEE Trans Pattern Anal Mach Intell* 2023.
- [60] Qiao W, Huang E, Zhang M, et al. Risk influencing factors on the consequence of waterborne transportation accidents in China (2013–2023) based on data-driven machine learning[J]. *Reliability Engineering & System Safety* 2025:110829.
- [61] Sun J, Zhong G, Huang K, et al. Banzhaf random forests: cooperative game theory based random forests with consistency[J]. *Neural Netw* 2018;106:20–9.

- [62] Egrioglu E, Yolcu U, Bas E, et al. Median-Pi artificial neural network for forecasting [J]. *Neural Comput Appl* 2019;31:307–16.
- [63] Tan Q, Fu M, Wang Z, et al. A real-time early warning classification method for natural gas leakage based on random forest[J]. *Reliability Engineering & System Safety* 2024;251:110372.
- [64] Wang H, Cai W, Wang Y. Optimization of a hybrid ejector air conditioning system with PSOGA[J]. *Appl Therm Eng* 2017;112:1474–86.
- [65] Van Huynh T, Tangaramvong S, Do B, et al. Sequential most probable point update combining gaussian process and comprehensive learning PSO for structural reliability-based design optimization[J]. *Reliability Engineering & System Safety* 2023;235:109164.
- [66] Zhong X, Liu R. Identifying critical nodes in interdependent networks by GA-XGBoost[J]. *Reliability Engineering & System Safety* 2024;251:110384.
- [67] Garg H. A hybrid PSO-GA algorithm for constrained optimization problems[J]. *Applied Mathematics and Computation* 2016;274:292–305.
- [68] Jamali B, Rasekh M, Jamadi F, et al. Using PSO-GA algorithm for training artificial neural network to forecast solar space heating system parameters[J]. *Applied Thermal Engineering* 2019;147:647–60.
- [69] Lu P, Mengdi L, Zhuo Z. The crowd dynamics under terrorist attacks revealed by simulations of three-dimensional agents[J]. *Artif Intell Rev* 2023;56(11): 13103–25.
- [70] Ran X, Liu Y, Pan H Y, et al. Shrimp phenotypic data extraction and growth abnormality identification method based on instance segmentation[J]. *Computers and Electronics in Agriculture* 2025;229:109701.
- [71] Wu Y, Schulz LE, Frank MC, et al. Emotion as information in early social learning [J]. *Curr Dir Psychol Sci* 2021;30(6):468–75.
- [72] Lu P, Wen F, Li Y, et al. Individual behaviors, social learning, and swarm intelligence: real case and counterfactuals[J]. *Expert Systems with Applications* 2022;207:117878.
- [73] Speiser JL, Miller ME, Tooze J, et al. A comparison of random forest variable selection methods for classification prediction modeling[J]. *Expert Syst Appl* 2019; 134:93–101.
- [74] Probst P, Boulesteix AL. To tune or not to tune the number of trees in random forest [J]. *J Mach Learn Res* 2018;18(181):1–18.
- [75] Bai J, Li Y, Li J, et al. Multinomial random forest[J]. *Pattern Recognit* 2022;122: 108331.
- [76] Aziz M R, Alfoudi A S. Feature selection of the anomaly network intrusion detection based on restoration particle swarm optimization[J]. *International Journal of Intelligent Engineering & Systems* 2022;15(5).
- [77] Amirteimoori A, Mahdavi I, Solimanpur M, et al. A parallel hybrid PSO-GA algorithm for the flexible flow-shop scheduling with transportation[J]. *Computers & Industrial Engineering* 2022;173:108672.